

SAVITRIBAI PHULE PUNE UNIVERSITY



A MINI PROJECT REPORT ON

“Twitter Sentiment Analysis using Python”

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WARJE, PUNE – 411058

2025 - 26



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CERTIFICATE

This is to certify that the project report entitled

“Twitter Sentiment Analysis using Python”

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is a bona fide work carried out by them under the supervision of **Ms. Jyoti Ragthawan** and it is submitted towards the partial fulfilment of the requirement of Savitribai Phule Pune University for Third Year.

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Certificate by Guide

This is to certify that **Aryan Jairam Shende** has completed the MINI Project work under my guidance and supervision and that, I have verified the work for its originality in documentation, problem statement, implementation and results presented in the Project. Any reproduction of other necessary work is with the prior permission and has given due ownership and included in the references.

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PROBLEM STATEMENT

The rapid growth of social media platforms like Twitter has led to the generation of large amounts of textual data containing users' opinions and sentiments. Analyzing this unstructured data manually is difficult and time-consuming.

The objective of this project is to perform sentiment analysis on Twitter data using Natural Language Processing (NLP) techniques. The tweet data is collected and preprocessed to remove noise such as special characters, URLs, and unwanted text. Using the VADER sentiment analyzer, each tweet is assigned a sentiment score and classified into positive, negative, or neutral categories.

Finally, the results are visualized using graphs to understand sentiment trends over time. This helps in extracting meaningful insights from social media data.

OBJECTIVES:

1. To collect and load Twitter dataset for analysis.
2. To preprocess and clean the textual data by removing noise such as special characters, URLs, and unwanted words.
3. To apply Natural Language Processing (NLP) techniques for analyzing text data.
4. To use the VADER sentiment analyzer to calculate sentiment scores of tweets.
5. To classify tweets into positive, negative, and neutral categories based on sentiment scores.
6. To visualize sentiment trends using graphical representations for better understanding.
7. To interpret the results and derive meaningful insights from the data.

INTRODUCTION

In today's digital world, social media platforms play a very important role in expressing opinions and sharing information. Among these platforms, Twitter is one of the most popular micro blogging services where users post short messages known as tweets. These tweets often reflect people's thoughts, emotions, and opinions about various topics such as products, services, events, and current trends.

With the continuous increase in the number of users, a huge amount of textual data is generated on Twitter every day. This data can be very useful for businesses, organizations, and researchers to understand public sentiment and make better decisions. However, manually analyzing such a large amount of unstructured data is difficult and inefficient.

To overcome this problem, **Sentiment Analysis** is used. Sentiment analysis is a technique of Natural Language Processing (NLP) that helps in identifying and extracting opinions from text data. It determines whether a given piece of text expresses a positive, negative, or neutral sentiment.

In this project, we perform sentiment analysis on Twitter data using Python. The dataset containing tweets is first preprocessed to remove noise such as special characters, URLs, and irrelevant words. Then, the **VADER (Valence Aware Dictionary and sEntiment Reasoner)** sentiment analysis tool from the NLTK library is used to calculate sentiment scores.

Based on these scores, tweets are classified into three categories:

- Positive
- Negative
- Neutral

Finally, the results are visualized using graphs to understand sentiment trends over time. This visualization helps in identifying patterns and gaining insights into public opinion.

This project demonstrates how data analytics and NLP techniques can be used to extract meaningful information from social media data and can be applied in real-world scenarios such as customer feedback analysis, brand monitoring, and market research.

REQUIREMENTS

Hardware Requirements:

1. **Processor:** Intel i3 or higher
2. **RAM:** Minimum 4 GB (8 GB recommended)
3. **Storage:** At least 500 MB free disk space
4. **System Type:** 64-bit computer/laptop
5. **Input Devices:** Keyboard and Mouse

Software Requirements:

1. **Operating System:** Windows 10/11 or Linux or macOS
2. **Programming Language:** Python 3.x
3. **Development Environment:**
 - Jupyter Notebook OR
 - Google Colab
4. **Python Libraries:**
 - **pandas** → for data handling
 - **nlTK** → for sentiment analysis
 - **plotly** → for visualization
 - **re (Regular Expressions)** → for text cleaning
5. **Browser (if using Colab):**
 - Google Chrome / Edge

ALGORITHM

Twitter Sentiment Analysis using NLP

1. Start

2. Import Required Libraries

- Import pandas, numpy, nltk, re, and plotly

3. Load Dataset

- Read CSV file containing Twitter data into DataFrame

4. Data Understanding

- Display dataset using head()
- Check structure using info()

5. Data Preprocessing

- Remove unwanted text such as:
 - URLs
 - User mentions (@username)
 - Hashtags and special characters
- Perform tokenization
- Filter meaningful words using NLTK corpus
- Store cleaned tweets in a new column

6. Download Required NLP Resources

- Download VADER lexicon
- Initialize SentimentIntensityAnalyzer

7. Sentiment Score Calculation

- Apply sentiment analyzer to each cleaned tweet
- Extract compound sentiment score

8. Sentiment Classification

- Assign labels based on score:
 - Positive $\rightarrow$ score > 0
 - Neutral $\rightarrow$ score $= 0$
 - Negative $\rightarrow$ score < 0

9. Feature Enhancement

- Add new columns for sentiment score and category

10. Data Aggregation

- Group tweets by date
- Count number of positive and negative tweets

11. Data Visualization

- Plot line graph using Plotly
- Represent sentiment trends over time

12. Result Analysis

- Interpret the graph to understand sentiment patterns

13. Stop

CODE

Import required libraries

```
import pandas as pd
import re
import nltk
from nltk.sentiment.vader import SentimentIntensityAnalyzer
import plotly.graph_objs as go
```

Load dataset

```
df = pd.read_csv('twitter_dataset_6432.csv')
```

Display dataset

```
print(df.head())
print(df.info())
```

Download NLTK resources

```
nltk.download('vader_lexicon')
nltk.download('words')
```

Create word set for cleaning

```
words = set(nltk.corpus.words.words())
```

Function to clean tweets

```
def cleaner(tweet):
    tweet = re.sub("@[A-Za-z0-9]+", "", str(tweet))
    tweet = re.sub(r"(?:\@|http?\://|https?\://|www)\S+", "", tweet)
    tweet = " ".join(tweet.split())
    tweet = tweet.replace("#", "").replace("_", " ")

    tweet = " ".join(w for w in nltk.wordpunct_tokenize(tweet)
                     if w.lower() in words or not w.isalpha())
    return tweet
```

Apply cleaning

```
df['tweet_clean'] = df['tweet'].apply(cleaner)
```

Initialize sentiment analyzer

```
sid = SentimentIntensityAnalyzer()
```

Calculate sentiment scores

```
df['sentiment'] = df['tweet_clean'].apply(lambda x: sid.polarity_scores(str(x))['compound'])
```

Classify sentiment

```
def sentiment_category(x):
```

```
    if x > 0:
```

```
        return 'positive'
```

```
    elif x == 0:
```

```
        return 'neutral'
```

```
    else:
```

```
        return 'negative'
```

```
df['sentiment_category'] = df['sentiment'].apply(sentiment_category)
```

Display processed data

```
print(df[['tweet', 'tweet_clean', 'sentiment', 'sentiment_category']].head())
```

Group data for visualization

```
positive = df[df['sentiment_category'] == 'positive']
```

```
negative = df[df['sentiment_category'] == 'negative']
```

```
positive = positive.groupby('date', as_index=False).count()
```

```
negative = negative.groupby('date', as_index=False).count()
```

Plot graph

```
fig = go.Figure()
```

```
fig.add_trace(go.Scatter(  
    x=positive['date'],  
    y=positive['id'],  
    mode='lines+markers',  
    name='Positive',  
    line=dict(color='green')  
))
```

```
fig.add_trace(go.Scatter(  
    x=negative['date'],
```

```
y=negative['id'],  
mode='lines+markers',  
name='Negative',  
line=dict(color='red')  
)
```

```
# Show graph
```

```
fig.show()
```

CONCLUSION

In this mini project, we successfully performed sentiment analysis on Twitter data using Python and Natural Language Processing (NLP) techniques. The dataset containing tweets was preprocessed to remove unwanted elements such as special characters, URLs, and irrelevant text.

Using the VADER sentiment analyzer from the NLTK library, each tweet was assigned a sentiment score and classified into positive, negative, and neutral categories. The processed data was then visualized using graphs, which helped in understanding sentiment trends over time.

The results demonstrate that sentiment analysis is an effective technique for extracting meaningful insights from social media data. This approach can be applied in real-world scenarios such as customer feedback analysis, product reviews, and decision-making processes.

REFERENCES

- NLTK Official Documentation – <https://www.nltk.org>
- Pandas Official Documentation – <https://pandas.pydata.org>
- Plotly Python Graphing Library – <https://plotly.com/python/>
- Python Official Website – <https://www.python.org>
- VADER Sentiment Analysis (NLTK) –
<https://www.nltk.org/api/nltk.sentiment.vader.html>
- Kaggle Datasets (Twitter Sentiment Data) – <https://www.kaggle.com>
- Stack Overflow (Programming Help & Debugging) – <https://stackoverflow.com>
- Practical Assignment PDF (Group C)